Yewei Huang

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Education

Doctor of Philosophy, Mechanical Engineering
Stevens Institute of Technology, NJ — 08/2019 - 05/2025

Master of Science, Surveying Engineering
Tongji University, Shanghai, China — 09/2016 - 03/2019

Bachelor of Science, Geographic Information System
Tongji University, Shanghai, China — 09/2012 - 06/2016

Positions

Post-doctoral Research Fellow, Dartmouth Reality and Robotics Lab
Dartmouth College, NH — 07/2025 - present

Graduate Student Research Assistant, Robust Field Autonomy Laboratory
Stevens Institute of Technology, NJ — 08/2019 - 05/2025

Skills

C++, Python, ROS, reinforcement learning, computer vision, Matlab, LaTex, Git.

Project 1 - Distributed Multi-Robot Perception and Navigation

Robust Field Autonomy Laboratory, Stevens Institute of Technology, NJ — Advisor: Dr. Brendan Englot

Distributed Multi-Robot LiDAR SLAM — Jan 2020 - Sep 2021

- Built a distributed multi-robot SLAM framework for robot teams equipped with lidar and inertial sensors.
- Github Link: https://github.com/RobustFieldAutonomyLab/DiSCo-SLAM.git
- Manuscript published in ***IEEE Robotics and Automation Letters***.

Multi-Robot Autonomous Exploration — Jun 2023 - May 2024

- Proposed a multi-robot autonomous exploration Algorithm using Expectation Maximization algorithm.
- Github Link: https://github.com/RobustFieldAutonomyLab/Multi-Robot-EM-Exploration.git
- Design a multi-robot exploration framework with Graph Neural Network.
- Manuscript published in ICRA'24.

Multi-Robot SLAM for Underwater Unmanned Vehicle (UUV)s — September 2024 - March 2025

- Built a distributed graph matching based multi-robot SLAM module for underwater robot teams.
- Manuscript accepted by IROS'25.

Project 2 - DARPA Symbiotic Design for Cyber Physical Systems Challenge

Robust Field Autonomy Laboratory, Stevens Institute of Technology, NJ — Advisor: Dr. Brendan Englot

Unmanned Aerial Vehicle (UAV) Flight Dynamics Design Aug 2021 - Aug 2022

- Designed a UAV flight dynamics parameter optimizer using multi-objective optimization.

Gaussian Process Motion Planning for UUVs Aug 2022 - May 2023

- Implement a Gaussian Process Motion Planning Algorithm for 3D underwater vehicle near seafloor.
- Github Link: https://github.com/RobustFieldAutonomyLab/Mission-Oriented-GP-Motion-Planning.git
- Manuscript published in ***IEEE Robotics and Automation Letters***.

Project 3 - IARPA Walk-Through Rendering from Images of Varying Altitude

Robust Field Autonomy Laboratory, Stevens Institute of Technology, NJ Advisor: Dr. Brendan Englot

Structure From Motion(SFM) with Varying Altitude Images Jan 2024 - Apr 2024

- Build and test the Structure From Motion pipeline for varying altitude images.
- Manuscript accepted by IROS'25.

Project 4 - SUWAC Human Robot Interaction

Vision and Robotics Laboratory, SRI International, NJ Advisor: Dr. Han-Pang Chiu

Multi-robot Scene Graph SLAM May 2024 - Aug 2024

- Built a distributed multi-robot SLAM with scene graph module for team of legged robots equipped with lidar, visual and inertial sensors.
- Manuscript preparing for submission.

Publications

1. **Y. Huang**, H. Wang, K. Zhan, J. Zhao, P. Gui, and T. Feng "Image-based localization for indoor environment using mobile phone" *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 40:211–215, 2015.
2. **Y. Huang**, J. Zhao, X. He, S. Zhang, and T. Feng "Vision-based semantic mapping and localization for autonomous indoor parking" *Proceedings of the IEEE Intelligent Vehicles Symposium (IV)*, 636–641, 2018.
3. **Y. Huang**, T. Shan, F. Chen and B. Englot "DiSCo-SLAM: Distributed scan context-enabled multi-robot lidar slam with two-stage global-local graph optimization" *IEEE Robotics and Automation Letters (RA-L)*, 7(2):1150–1157, 2021.
4. **Y. Huang**, X. Lin, M. Hernandez-Rocha, S. Narain, K. Pochiraju and B. Englot "Mission-oriented Gaussian Process Motion Planning for UUVs over Complex Seafloor Terrain and Current Flows" *IEEE Robotics and Automation Letters (RA-L)*,9(2):1780-1787, 2024.
5. **Y. Huang**, X. Lin, and B. Englot "Multi-Robot Autonomous Exploration and Mapping Under Localization Uncertainty with Expectation-Maximization" *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 7236–7242, 2024.
6. **Y. Huang**, J. McConnell, X. Lin, and B. Englot "DRACo-SLAM2: Distributed Robust Acoustic Communication-efficient SLAM for Imaging Sonar Equipped Underwater Robot Teams with Object Graph Matching" *Submitted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, accepted, to appear, 2025.
7. J. Zhao, **Y. Huang**, X. He, S. Zhang, C. Ye, T. Feng, and L. Xiong "Visual semantic landmark-based robust mapping and localization for autonomous indoor parking" *Sensors*, 19(1):161, 2019.

8. F. Chen, J. Martin, **Y. Huang**, J. Wang, and B. Englot "Autonomous exploration under uncertainty via deep reinforcement learning on graphs" *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 6140–6147, 2020.
9. F. Chen, P. Szenher, **Y. Huang**, J. Wang, T. Shan, B. Shi and B. Englot "Zero-shot reinforcement learning on graphs for autonomous exploration under uncertainty" *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 5193–5199, 2021.
10. J. McConnell, **Y. Huang**, P. Szenher, I. Collado-Gonzalez and B. Englot "DRACo-SLAM: Distributed Robust Acoustic Communication-efficient SLAM for Imaging Sonar Equipped Underwater Robot Teams" *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 8457–8464, 2022.
11. J. Wang, F. Chen, **Y. Huang**, J. McConnell, T. Shan, and B. Englot "Virtual maps for autonomous exploration of cluttered underwater environments" *IEEE Journal of Oceanic Engineering*, 47(4):916–935, 2022.
12. X. Lin, **Y. Huang**, D. Sun, T.-Y. Lin, B. Englot, R. M. Eustice, and M. Ghaffari "A Robust Keyframe-Based Visual SLAM for RGB-D Cameras in Challenging Scenarios" *IEEE Access*, 11, 97239–97249, 2023.
13. X. Lin, P. Szenher, **Y. Huang**, and B. Englot "Distributional Reinforcement Learning Based Integrated Decision Making and Control for Autonomous Surface Vehicles" *IEEE Robotics and Automation Letters (RA-L)*, 2024.
14. K. Doherty, A. Papalia, **Y. Huang**, D. Rosen, B. Englot, and J. Leonard "MAC: Maximizing Algebraic Connectivity for Graph Sparsification" *arXiv e-prints*, arXiv–2403, 2024.
15. Y. Li, **Y. Huang**, B. Gaudel, H. Jafarnejadsani, and B. Englot "CVD-SfM: A Cross-View Deep Front-end Structure-from-Motion System for Sparse Localization in Multi-Altitude Scenes" *Submitted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, accepted, to appear, 2025.
16. Y. Li, **Y. Huang**, W. Gaozhang, and Y. Liu "Autonomous Exploration with Virtual Map using Unmanned Surface Vehicles" *Submitted to IEEE International Conference on Automation and Computing (ICAC)*, accepted, to appear, 2025.

Mentoring

I have mentored the following undergraduate and master's students with my doctoral advisor, Prof. Brendan Englot:

- **Thomas Roff**. "Optimization of Utility Function Weights for Active Autonomous Exploration and Mapping With Multi-robot Systems" *Master's thesis*, Stevens Institute of Technology, 2024.
- **Sneha Banda**, **Jesse Knuckles**. "Modular Sensor Fusion Package for Localization" *Undergraduate Senior Design*, sponsored by Kearfott Guidance and Navigation, Stevens Institute of Technology, 2025.

I have mentored the following doctoral student in collaboration with Prof. Yuanchang Liu from UCL:

- **Ye Li**. "Autonomous Exploration with Virtual Map using Unmanned Surface Vehicles" *The 30th International Conference on Automation and Computing (ICAC)*, 2025.

Awards and Honors

- Academic Year 2019-2020 Stevens Provost Doctoral Fellowship
- Academic Year 2024-2025 Stevens Excellence Doctoral Fellowship
- Academic Year 2024-2025 Paul Kaplan Award

Reviewer

- IEEE Transactions on Robotics (T-RO).
- International Journal of Robotics Research (IJRR).
- IEEE Robotics and Automation Letters (RA-L).
- Journal of Field Robotics (JFR).
- Elsevier Applied Ocean Research.
- IEEE International Conference on Robotics and Automation (ICRA).
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

Seminars

- Future Leaders in Robotics and AI Seminar hosted by Maryland Robotics Center on March 28, 2025.